Ablation Design Reading Synthesis

Research reading artifact | fictional/internal | 2026-07-13

Analytical frame. This review frames ablation as a causal hygiene exercise inside a fictional Helix Forge Lab setting. The fixed point is run R184 on 2026-07-13, where Model Alpha M-14 reaches F1 0.873 on the synthetic five-label evaluation; the note does not infer real-world causality.

Source-by-design matrix

Each synthetic/internal source proposes a testable removal: context features, confidence features, or abstention rule. The matrix prevents a vague 'simplified model' comparison.

Interaction and confounding

Removing a feature family can alter calibration and class routing together. An apparent gain may reflect a changed threshold rather than the removed representation.

Ablation reporting template

Record baseline, single intervention, frozen evaluation slice, metric panel, observed delta, and interpretation boundary in that order.

Working matrix

Intervention	Confound to control	Report field
Remove context cues	Routing fallback	Per-label recall
Remove confidence cues	Threshold setting	Reliability bins
Change abstention rule	Class prevalence	Precision / recall

Source-by-design citation matrix

Synthetic/internal source	Proposed intervention	Evidence expected
Single-Factor Ablation Protocol	Remove context cues	Route/resolve recall change
Confounding Checklist	Hold threshold fixed	Calibration does not explain delta
M-14 Intervention Register	Change abstention rule	Precision-recall boundary shifts

Confound audit

1. Was exactly one feature family changed?
2. Did the evaluation slice and threshold remain fixed?
3. Could the observed delta be caused by routing fallback?
4. Is a label-level result available to challenge the aggregate claim?

Decision boundary

An ablation conclusion is provisional until the audit answers each question with a recorded artifact.